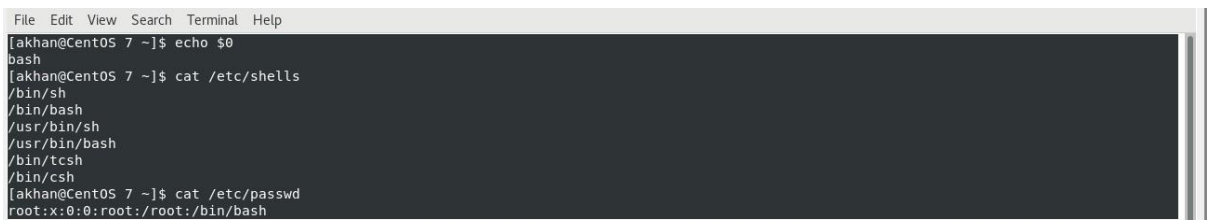


What is Shell?

- > It's like a container
- > Interface between user and Kernel/OS
- > CLI is a Shell or GUI is a Shell

Find your Shell:

- > echo \$0
- > Available Shells "cat /etc/shells"
- > your Shell? "cat etc/passwd"
- > Windows GUI is a Shell
- > Linux KDE GUI is Shell
- > Linux sh, bash etc. is a shell



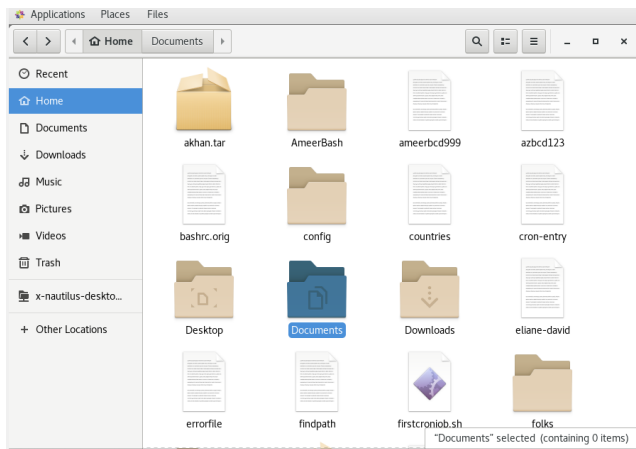
```
File Edit View Search Terminal Help
[akhan@CentOS 7 ~]$ echo $0
bash
[akhan@CentOS 7 ~]$ cat /etc/shells
/bin/sh
/bin/bash
/usr/bin/sh
/usr/bin/bash
/bin/tcsh
/bin/csh
[akhan@CentOS 7 ~]$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
```

Shell Script:

Shell script is nothing but putting instructions in a sequential order like do this, do that and Run it.

Types of Shells:

1. **Gnome and KDE:** These are the GUI Shells within Linux and you can opt while installing.



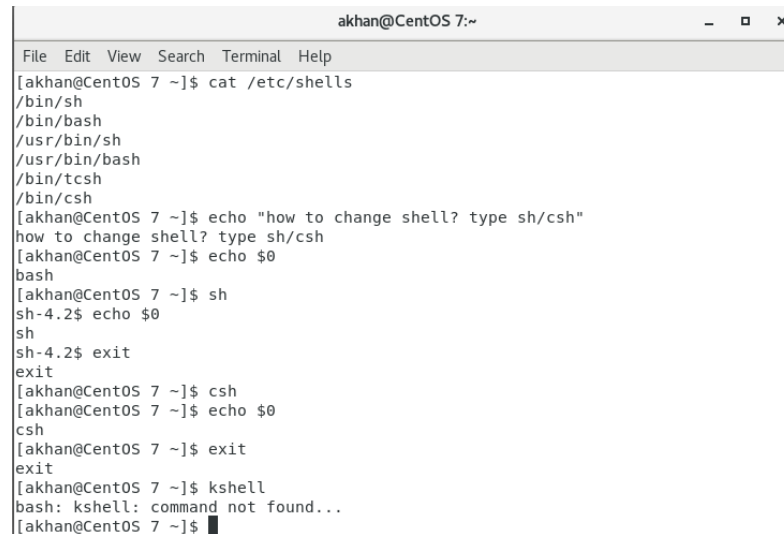
2. **sh and bash:** These are Linux/Unix based CLI shells and bash shell has some superiority than sh shell such auto completion tab, history saving, aliases and Advanced Arrays and Maths.

```
akhan@CentOS 7:~  
File Edit View Search Terminal Help  
[akhan@CentOS 7 ~]$ ls -ltr |tail -10  
-rw-rw-r--. 1 akhan akhan      5 Apr 11 2022 Russia  
-rw-rw-r--. 1 akhan akhan      27 Apr 11 2022 sep2  
-rw-rw-r--. 1 akhan akhan 7823360 Feb  4 2024 akhan.tar  
-rw-rw-r--. 1 akhan akhan 15646720 Feb  4 2024 linuxcourse.tar  
drwxrwxr-x. 2 akhan akhan      6 Feb  4 2024 unzip  
-rw-rw-r--. 1 akhan akhan      47 Sep 27 2025 firstcronjob.sh  
-rwxrwx-rw-. 1 akhan akhan      72 Sep 27 2025 testcron.sh  
-rwxr--r--. 1 akhan akhan     1802 Sep 27 2025 cron-entry  
-rw-rw-r--. 1 akhan akhan      26 Sep 27 2025 rsynctestfile.txt  
drwx-----T. 5 akhan akhan    4096 Oct 10 2025 linuxcourse  
[akhan@CentOS 7 ~]$
```

3. **cshell or tcshell:** Need to be proficient in c or c++ to use this.
4. **kshell:** It will use in Solaris most of the time and kshell improves on bash by adding floating point, auto job control etc.

Note: It is important to understand about diff shells as all scripts are not compatible in each other.

How to change shells?



```
akhan@CentOS 7:~  
File Edit View Search Terminal Help  
[akhan@CentOS 7 ~]$ cat /etc/shells  
/bin/sh  
/bin/bash  
/usr/bin/sh  
/usr/bin/bash  
/bin/tcsh  
/bin/csh  
[akhan@CentOS 7 ~]$ echo "how to change shell? type sh/csh"  
how to change shell? type sh/csh  
[akhan@CentOS 7 ~]$ echo $0  
bash  
[akhan@CentOS 7 ~]$ sh  
sh-4.2$ echo $0  
sh  
sh-4.2$ exit  
exit  
[akhan@CentOS 7 ~]$ csh  
[akhan@CentOS 7 ~]$ echo $0  
csh  
[akhan@CentOS 7 ~]$ exit  
exit  
[akhan@CentOS 7 ~]$ kshell  
bash: kshell: command not found...  
[akhan@CentOS 7 ~]$
```

You need to make sure what is the environment before you executing the script because if you written script in kshell and trying to run in bash shell than it will not work.

How to run shell scripts?

1. There are two paths, absolute and relative path.

- ➔ Absolute: `sh /home/akhan/Scripts/myscript.sh`
- ➔ Relative path: `cd /home/akhan/Scripts`
`sh ./myscript.sh`

```
[akhan@CentOS 7 Scripts]$ cd /  
[akhan@CentOS 7 /]$ pwd  
/  
[akhan@CentOS 7 /]$ sh /home/akhan/Scripts/myscript.sh  
Hello World from path: /  
[akhan@CentOS 7 /]$ cd /home/akhan/Scripts/  
[akhan@CentOS 7 Scripts]$ pwd  
/home/akhan/Scripts  
[akhan@CentOS 7 Scripts]$ sh ./myscript.sh  
Hello World from path: /home/akhan/Scripts  
[akhan@CentOS 7 Scripts]$
```